

Jaco du Toit

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SUMMARY

I am an AI safety research engineer with a background in ML research and engineering. My work currently focuses on designing evaluations for frontier LLM-based agents to measure dangerous capabilities. I am excited by the rapid advancements in AI but concerned about the risks they pose, and I am committed to helping mitigate them.

EXPERIENCE

AI Safety South Africa

Apr 2025 – Present

Research Engineer

Research focused on measuring the adaptability of LLM agents as a proxy for real-world performance, in collaboration with the Science of Evaluations team at the UK AISI and funded by the UK AISI Challenge Fund. Role spans experimental design, implementation, empirical analysis, and writing.

ClearObject

Apr 2025 – Present

Machine Learning Engineer

Work on deep learning projects applying computer vision to safety-critical monitoring in real-world environments, such as infrastructure and industrial sites.

AI Safety Engineering Taskforce

Jan 2025 – Apr 2025

Research Engineer

Designed and implemented evaluations for frontier LLMs as part of the UK AISI bounty program, focusing on autonomous self-improvement, situational awareness, and sandbagging.

Matogen AI

May 2024 – Apr 2025

Machine Learning Engineer

Delivered data science pipelines and deep learning solutions for leading financial, agricultural, and retail institutions in Africa.

EDUCATION

MSc Computer Science

Jan 2023 – Mar 2024

Stellenbosch University • Cum Laude (85%)

Thesis: Hierarchical Text Classification with Transformer-based Language Models.

BSc (Hons) Computer Science

Jan 2022 – Dec 2022

Stellenbosch University • Cum Laude (86%)

Focus on machine learning, with a mini-thesis in machine learning and natural language processing.

BCom Mathematical Sciences (Data Science)

Jan 2019 – Dec 2021

Stellenbosch University • Cum Laude (87%)

Majors in Mathematics and Computer Science.

SKILLS

Programming Languages

Experienced: Python | Familiar: C | Java | R | JavaScript | SQL

Frameworks & Libraries

PyTorch | TensorFlow | HuggingFace | Inspect | Scikit-learn | Pandas | NumPy | PySpark | OpenCV | PIL | ONNX | Matplotlib | Seaborn

Software Development

Git | Docker | CLI | Agile

Cloud Platforms

Google Cloud Platform | Amazon Web Services

PUBLICATIONS

- Brown, S. F., **du Toit, J.**, Hyams, L., and Anisimov, D. (2025). *Precursors, Proxies, and Predictive Models for Long-Horizon Tasks*. NeurIPS 2025 LLM Evaluation Workshop Poster.
- **du Toit, J.** and Dunaiski, M. (2024). *Prompt Tuning Discriminative Language Models for Hierarchical Text Classification*. Natural Language Processing.
- **du Toit, J.**, Redelinghuys, H., and Dunaiski, M. (2024). *Introducing Three New Benchmark Datasets for Hierarchical Text Classification*. arXiv:2411.19119.
- **du Toit, J.** and Dunaiski, M. (2023). *Hierarchical Text Classification Using Language Models with Global Label-Wise Attention Mechanisms*. In: Artificial Intelligence Research. SACAIR 2023. CCIS vol. 1976, Springer, Cham.

AWARDS, HONOURS & SCHOLARSHIPS

Scholarship for Master’s Studies	2023 – 2024
School for Data Science and Computational Thinking.	
Best Presentation Award	2023
Stellenbosch University, Faculty of Science Postgraduate Research Conference.	
Rector’s Award for Academic Excellence	2022
Stellenbosch University – awarded for being in the top 10 of the faculty.	
Academic Merit Award	2019 – 2021
Stellenbosch University – awarded for being in the top 5% of the faculty.	
Golden Key International Honour Society	2019
Awarded for being in the top 15% of the class.	

ADDITIONAL EDUCATION

ML4Good – Teaching Assistant	Apr 2026
Teaching assistant for the ML4Good AI safety bootcamp.	
Cooperative AI Summer School	Jul 2025
Week-long residential summer school on cooperative AI research.	
Introduction to Cooperative AI	Apr 2025 – Jun 2025
Cooperative AI Foundation – 3-month part-time course with research project in final month.	
AI Safety, Ethics, and Society Course	Feb 2025 – Apr 2025
Center for AI Safety – 3-month part-time course with research project in final month.	
AI Safety Camp Cape Town	Nov 2024
Condor Initiative – one-week upskilling course in AI safety.	
AI Safety Fundamentals Alignment Course	Oct 2024 – Feb 2025
BlueDot Impact – 3-month part-time course with research project in final month.	